Quantum CubeSat (QubeSat)

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Introduction/Abstract

QubeSat's primary goal is to research the effects of space environments on quantum gyroscopes based on NV– centers in diamond. NV– centers are defect points in the diamond lattice, where a neighboring pair of lattice positions consists of a nitrogen atom, and a lattice vacancy. This structure is useful because its spin state can be read out using laser excitation and photodiode fluorescence, facilitating angular motion measurements. NV– centers in diamond can also be used to create magnetometers and other sensors, which are also critical in space missions; this project could provide a platform for further qualification of these sensors. Space flight is necessary for this testing because there are many different factors including temperature, radiation, and magnetic field variation that could affect the performance of the quantum gyroscope. Therefore, we propose a proof of concept for the feasibility of NV– diamond based quantum gyroscopes in Low-Earth Orbit conditions.

The QubeSat project will involve fabricating and testing the quantum gyroscope on Earth, and comparing its performance in space to our on-earth measurements. STAC will be building the quantum gyroscope and testing it against a MEMS-based gyroscope since it is the standard technology used in CubeSats. We will be gathering the parameters of both the quantum gyroscope and the MEMS gyroscope on Earth using methods described later in this poster, and comparing against these parameters while in space to determine if space conditions have a significant effect on our quantum gyroscope.

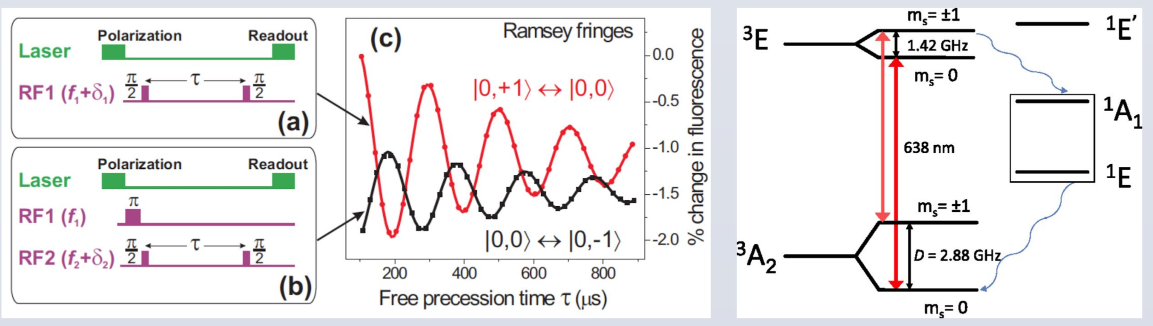
In comparison to more traditional onboard small scale gyroscopes, quantum gyroscopes could potentially provide an edge in several key aspects. In particular, the project seeks to demonstrate four key advantages of quantum gyroscopes in furthering space technology:

1. Improved Resolution: This device can improve the gyroscopic sensitivity by an order of magnitude (10-4 $rad/s/mm^3/Hz^{1/2}$ for MEMS gyroscope, to 10-5 $rad/s/mm^3/Hz^{1/2}$ for quantum gyroscope).
2. Improved Drift Stability: MEMS based gyroscopes suffer from a phenomenon known as drift, which decreases the accuracy and precision of the gyroscope over time. The quantum gyroscope promises to drastically reduce the drift error by at least two orders of magnitude, and provides a means to collect reliable data over longer periods of time,.
3. Single-Device Multiple-Sensor Integration: The ability of the quantum gyroscope to function as a versatile sensor such as a magnetometer would reduce the hardware requirements for satellites, and can improve gyroscope performance in environments with known magnetic field (such as Low-Earth Orbit) by providing a baseline for sensor calibration.
4. Temperature Operational Range: This device is stable over a wide temperature range, from cryogenic temperature to 600-700K.

Quantum Gyroscope Payload

The quantum gyroscope used in this experiment is largely based on the works of professors Dmitry Budker from UC Berkeley and Paola Cappellaro from the Massachusetts Institute of Technology,. The NV– centers can be detected using Optically Detected Magnetic Resonance and manipulated using optical and RF pulses tuned to the transition frequencies. In the proposed scheme, a rotation of the diamond lattice around the NV– center symmetry axis causes a Berry phase to be accumulated during the free precession of the NV- center. This shift can be measured through using Ramsey Interferometry. By using NV– centers with different orientations, the quantum gyroscope can be modified to serve as a 3-axis sensor.

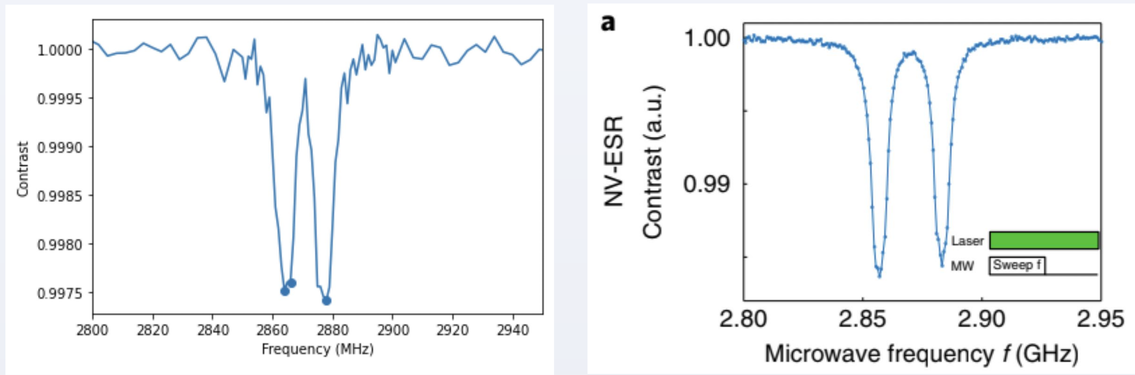
Gyroscope Operational Procedure:



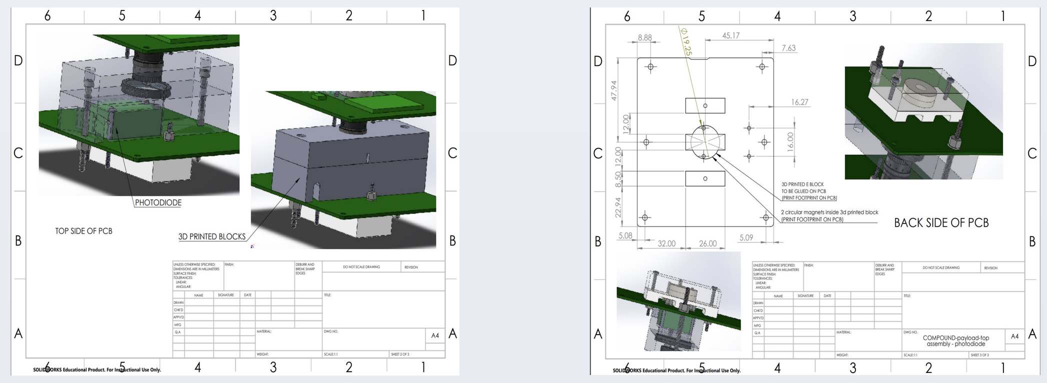
1. Initialization: All the spins in the lattice are optically polarized using a 532 nm laser.
2. *π*/2 Pulse: A microwave pulse is used to excite the spins into a superposition of the spin zero and plus or minus one spin states
3. Free Precession: The system is allowed to freely evolve within the background magnetic field, during which time a Berry phase is accumulated
4. *π*/2 Pulse: Another microwave pulse of the same duration is applied to collapse the superposition of spins into populations in the ground and excited states.
5. Readout: The sample is illuminated with a laser. The population of spins in the energy states is measured by collecting the fluorescence from the readout sequence using a photodiode array.

Continuous Wave Operation: Gyroscope sensitivity is based on the small shifts in the precession frequency due to the fictitious magnetic field introduced in the rotating reference frame. These shifts in the precession frequency lead to changes in measured intensity at the selected point in the Ramsey fringe. However, variation in environmental conditions (most notably the magnetic field environment and temperature) also lead to shifts in the precession frequency, introducing noise in the gyroscope measurement. In low-earth orbit environments with minimal device isolation, these sources of noise may be up to orders of magnitude larger than the target spin signal. The effects of environmental magnetic field and temperature variation are well-known and quantifiable, and can be post-processed out of the final signal given that the precise resonance frequencies are known. This can be accomplished through a continuous wave calibration, an ODMR processes in which the system is continuously optically pumped while the microwave-pumping frequency is swept. Resonances can be resolved when the fluorescence diminishes due to relaxation through the non-radiative sideband. Temperature and magnetic field information is a byproduct of this calibration process, and expected device noise can be quantified. For this reason, we include a continuous wave operational mode.

Continuous Wave Experimental Results:



Payload Structures

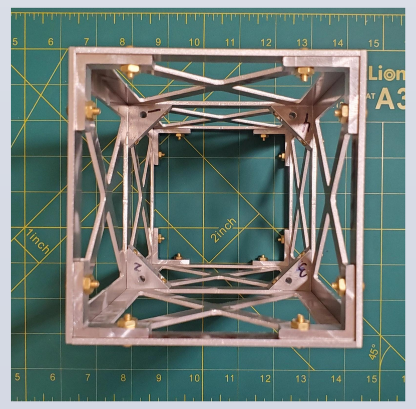


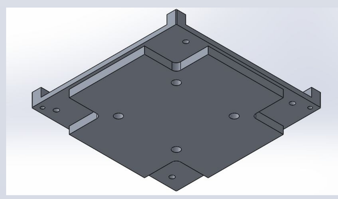
QubeSat's payload structure comprises of a laser, laser driver, laser socket, RF-MW generator, collimator, optical filter, laser diode, photodiode, laser PCB, and microcontroller. The diamond is placed on top of the optical filter and is equidistant from 2 pairs of magnets on either side, which are used to split the energy levels of the various NV center orientations. The laser, diamond, photodiode, magnets, optical filter, collimator, are all held in place with customized 3D printed structures. A microwave near-field loop antenna from the RF-MW generator is put on top the diamond.

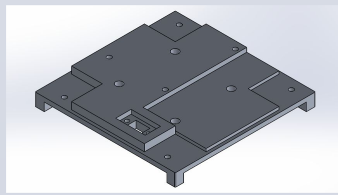
Structures

QubeSat's body is an assembly of four separate trussed 2mm-thick aluminum pieces joined at the corners by brackets. At each end is an aluminum cap with four venting holes to prevent the buildup of atmospheric pressure. Four aluminum rods run along the length of QubeSat. All PCBs were mounted on this set of rods. To create clearance between successive boards, hollow cylindrical aluminum spacers were used. All screws and nuts were fixed using Loctite and/or epoxy during final assembly.

Component Name	Material	Manufacturing Method	Anodized?
Body Truss Pieces	Aluminum	Laser Cutting	Yes
End Caps	Aluminum	CNC Machining (Protolabs)	Yes
PCB mounting rods	Aluminum	Hand Cut from Stock	No
Triangle Brackets	Aluminum	CNC Machining (Protolabs)	No
L Brackets	Aluminum	Hand Cut and Drilled	No
Spacers	Aluminum	Hand Cut from Stock	No

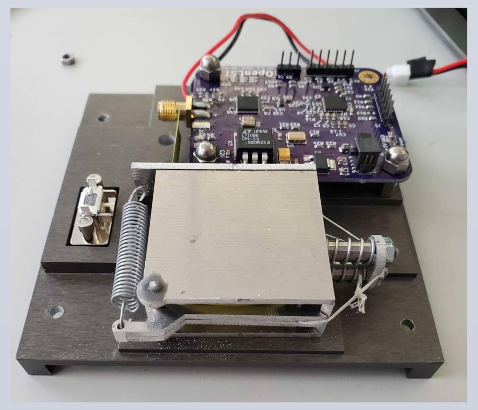






Antenna Deployment

QubeSat's antenna is a segment of measuring tape that is wound up inside an aluminum box which is coated on the inside with an electrically insulating film. This box is spring loaded and rigged with a burn wire release system, which opens a trapdoor and allows the measuring tape to unwind itself. The other end of the antenna slides through a slit in the aluminum box and is directly connected to the SMA port on the OpenLST board mounted directly adjacent to the antenna box.



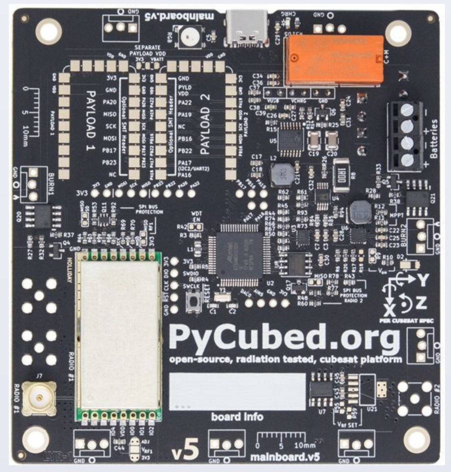
Power Systems + Communications

The Qubesat's power system is comprised of 16 Spectrolab XTJ Prime solar panels. Energy storage is handled by the PyCubed battery board which uses a 2S3P Li-Ion battery configuration and contains a number of built in protection features. The cells used by this board are 3.7V 18650 cells and are capable of powering the Qubesat for multiple orbits before needing to be recharged.

The Qubesat communicates with the ground station using the pycubed's built in HopeRF radio adjusted to work with FSK modulation at 437MHz. Custom modification of the underlying library was required to configure the radio was required to operate in this range. The communication scheme uses a simple packet structure with a built in checksum to detect corrupted packets.

Software

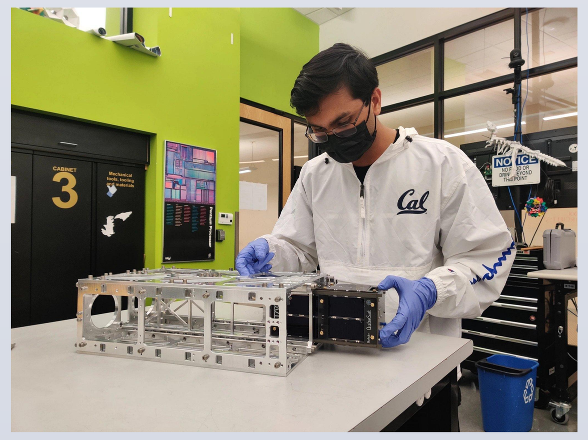
QubeSat uses the PyCubed board as its main board and an STM32 microcontroller as its payload board. Using a PyCubed allows us to use Python for our flight software. The PyCubed board triggers the experiment by turning on the laser and resetting the payload board. Once reset, the payload board collects data from the experiment and transmits it through UART back to the PyCubed. From there, the PyCubed queries its onboard IMU for telemetry data and writes both the experiment data and telemetry data to an SD card.

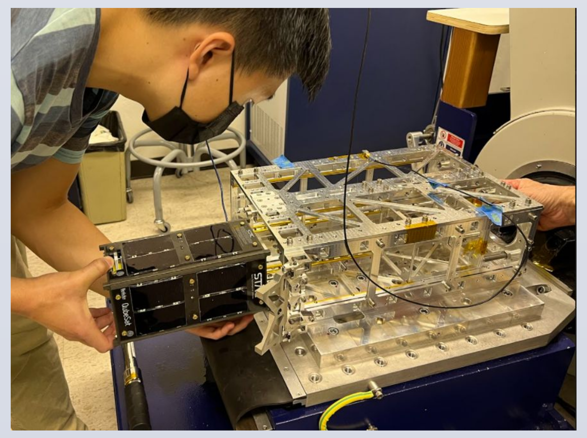


Testing

Fit Check: Prior to the vibration test, a fit check was performed with the dispenser to ensure that the QubeSat could easily be ejected during deployment since the rail system relied on exact dimensions as specified in the Interface Control Document (ICD).

Vibration Testing: Vibration testing was completed at SSL, the Space Sciences Laboratory at Berkeley. Qubesat was mounted onto the vibrational testing apparatus and run under three independent variables: Axis of vibration (XYZ), vibration frequency, and vibration amplitude. Inertial readings were taken to ensure no internal components had fallen apart. The first round of vibration testing was performed on October 8th. However, after the Z-axis test we found that the screws holding our magnetorquers in place had backed out and the test was not passed. A second vibration test was performed on November 2nd, after loctite was used to secure the magnetorquers as well as any other loose components. QubeSat passed the second vibration test, and was thus mechanically ready for launch. The next month and a half was spent on software, testing, and any last compliance documentation.





Integration + Launch

Preparing the QubeSat for the Astra VCLS Demo 2 launch took about one year from initial integration meetings to final delivery. Some of the external deliverables included passing a Fit Check and Vibration Test (described above), obtaining FCC and NOAA licenses, ensuring size, mass, magnetic field, etc. complied with Astra and other CubeSats on the mission. One of the major challenges in the last semester of the QubeSat project was integrating each of the separate subsystems together inside the QubeSat structure.

After completing the Vibration Test in November 2021, the main focus was on finalizing the software and testing each of the components. In December 2021, QubeSat was integrated into the dispenser for the Astra mission at their facility in Alameda, CA.

QubeSat launched with Astra in early February 2022. The launch was originally planned for the first weekend of February, and several QubeSat team members made the trip to Florida to watch the launch. After the launch was scrubbed twice earlier in the week, STAC members had the opportunity to go to the Astra facility in Alameda to watch the launch. Unfortunately, the mission was not a success due to some electrical and software issues, causing the payloads to be lost.

Conclusions

Over the two and a half years of this project, we learned a lot as a team and as a club. QubeSat was the first major launch for STAC in three years, and the first CubeSat ever launched by the club. This meant that almost all designs were entirely new, and we learned about the integration process as we went. Launching with a startup company like Astra was also a new experience, and it was exciting being a part of such a new technology demonstration.

Using open-source designs such as PyCubed and OpenLST for QubeSat was very cost effective and easy to modify compared to off-the-shelf (COTS) solutions, but in many cases the lack of information or support made development difficult. Especially with our comparative lack of experience in communications and power systems, we were very thankful for the help from our advisors and other student groups around the country who we got the opportunity to work with.

The Quantum Gyroscope as our payload was an ambitious choice because we would be taking a technology only proven in lab conditions, and then miniaturizing it, making it robust in changing temperature and magnetic field, and using significantly cheaper parts.

The majority of the project, besides the first and last semesters, was done virtually due to the COVID-19 pandemic. Doing hardware work while virtual was very difficult. Some progress was made by shipping the hardware to members' houses and working asynchronously. However, the majority of the progress was on the CAD and software side. Once UC Berkeley began to open up again for in-person in Fall 2021, the majority of assembly and hardware work began.

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